Derivatives :

$$f(x) = (3x^2+1)^4$$

$$f'(x) = 4(3x^2+1)^3 \cdot 6x = 24x(3x^2+1)^3$$

$$f(x) = \sqrt{x^2+3x}$$

$$\Rightarrow f(x) = (x^2+3x)^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2}(x^2+3x)^{-\frac{1}{2}} \cdot (2x+3)$$

$$f'(x) = \left(\frac{2x}{2} + \frac{3}{2}\right)(x^2+3x)^{-\frac{1}{2}}$$

$$f'(x) = \left(x+\frac{3}{2}\right)(x^2+3x)^{-\frac{1}{2}}$$

$$f(x) = \ln(x^2+3x+1)$$

$$f'(x) = \frac{2x+3}{x^2+3x+1}$$

Day 5